WELL-UP

Sales Performance Dashboard

Project Documentation

Executive Overview | Data-Driven Insights for Strategic Decision-Making

Prepared by: Chukwuemeka Jayfresh

Tool: Power BI Desktop | Dashboard Views: 6 | Date Range: Jan 2015 – Mar 2017

Table of Contents

1. Project Overview

This project delivers a comprehensive Sales Performance Dashboard for Well-Up, built in Power BI Desktop. The dashboard provides executive-level visibility into revenue trends, profit margins, regional performance, customer behavior, and product category contributions over the period January 2015 to March 2017.

The dashboard consists of six dedicated report pages — Overview, Sales Analysis, Product Analysis, Customer Insights, Time Intelligence, Geography, and DAX Metrics — enabling decision-makers to drill into performance at every level of the business.

Interactive filters including a Region Filter and a Date Range slicer are available across pages, allowing stakeholders to isolate specific markets and time periods for targeted analysis.

2. Business Introduction

Well-Up is a data-driven sales organization operating across multiple geographic regions including Australia, Canada, Central, France, Germany, Northeast, and the Southwest and Northwest territories of the United States. The company serves a diverse customer base spanning multiple occupational demographics and gender groups.

With a sales database updated as recently as May 21, 2024, Well-Up generates revenue across multiple product categories and relies on strategic analysis to drive profitability. The Sales Performance Dashboard was commissioned to bring together all key commercial metrics into a single, interactive reporting environment for executive and operational stakeholders.

3. Business Problem

Well-Up faced key challenges in tracking and interpreting its sales performance effectively across its multi-region, multi-category business:

Challenge	Description
Fragmented Reporting	Sales, profit, and return data were tracked in silos with no unified view across regions, products, and customer segments.
Limited Time Intelligence	The company lacked the ability to compare performance across years, quarters, and months dynamically, making trend identification difficult.

Regional Blind Spots	Revenue distribution across geographies was not visualized, making it hard to prioritize high-value markets and identify underperforming regions.
Customer Segmentation Gap	Without visibility into revenue by occupation and gender, marketing and sales teams could not tailor strategies to high-value customer groups.
Product Category Clarity	The contribution of individual product categories to overall revenue was not clearly quantified, leading to suboptimal inventory and promotion decisions.

4. Business Objectives

The primary objectives of this project are:

- **Unified Performance View**
  - Consolidate revenue, profit, quantity sold, and return data into a single executive dashboard with real-time filter capability.
- **Regional Analysis**
  - Visualize revenue performance by region to identify top markets and guide resource allocation.
- **Time Intelligence**
  - Enable year-on-year, quarter-on-quarter, and month-on-month trend analysis using DAX time intelligence functions.
- **Customer Segmentation**
  - Break down revenue by customer occupation and gender to support targeted sales and marketing decisions.
- **Product Category Insights**
  - Quantify the revenue contribution of each product category to optimize stock, pricing, and promotional strategy.
- **DAX Metrics Transparency**
  - Surface calculated measures (profit margin, growth rates, return ratios) in a dedicated metrics page for analytical transparency.

5. Tools Used

Tool	Purpose
Power Query	Data ingestion, cleaning, transformation, and preparation before loading into the Power BI model.

Power Query — Applied Steps	Step-by-step shaping: removing nulls, deduplication, column splitting, type changes, and encoding.
Power BI Desktop	Dashboard layout, report page design, and publication of interactive visuals.
Power BI — DAX	Custom measures for Total Revenue, Total Profit, Profit Margin %, Total Quantity Sold, Total Returns, and YoY Growth.
Power BI — Visuals	KPI cards, bar charts, donut charts, column charts, line/area charts, and scatter plots.
Power BI — Slicers	Region Filter buttons and Date Range slicer for dynamic cross-page filtering.
Power BI — Bookmarks	Navigation between dashboard pages (Overview, Sales Analysis, Product Analysis, etc.) via the left nav panel.

6. Data Preparation & Cleaning Process

Before building the dashboard, a structured data preparation process was performed in Power Query within Power BI Desktop to ensure the sales database was clean, consistent, and correctly typed for DAX calculations and visual accuracy.

6.1 Check for Missing Values

Purpose: Identify and resolve null or empty values across the sales dataset that could produce incorrect totals or blank visuals.

Step	Action	Details
1	**Load data into Power Query**	In Power BI Desktop, click 'Transform Data' to open the Power Query Editor with the sales database loaded.
2	**Enable Column Quality view**	Go to the View tab in Power Query and check 'Column Quality'. This displays Valid, Error, and Empty percentage bars below each column header.
3	**Identify columns with empty values**	Columns where Empty % is greater than 0 require attention. Click the column to preview affected rows in the data pane.
4	**Replace missing values**	Right-click the affected column > Replace Values. For categorical fields (e.g., Region, Gender) use 'Unknown'; for numeric fields use 0 or the column average.
5	**Close & Apply**	Click 'Close & Apply' to commit all replacements and reload the cleaned dataset into the Power BI model.

6.2 Find and Remove Duplicates

Purpose: Ensure each sales transaction is recorded only once, preventing inflated revenue or quantity figures in the dashboard.

Step	Action	Details
1	**Select the query**	In the Power Query Editor, select the primary sales transactions table from the Queries pane on the left.
2	**Identify key columns**	Determine the columns that together uniquely identify a record (e.g., Order ID, Customer ID, Product ID, Order Date).
3	**Remove duplicates**	Highlight the key columns using Ctrl + Click, then go to Home tab > Remove Rows > Remove Duplicates.
4	**Review row count**	Compare the row count shown in the status bar before and after the step to confirm how many duplicates were removed.
5	**Verify in Applied Steps**	Confirm the 'Removed Duplicates' step is recorded in the Applied Steps pane. Click it to preview the resulting table.

6.3 Data Transformation

Purpose: Format raw data into the correct structure required for time intelligence, regional grouping, and customer segmentation analysis.

Step	Action	Details
1	**Split date-time columns**	Select the timestamp column, then go to Transform > Split Column > By Delimiter (space) to separate the date and time into two columns.
2	**Set date column type**	Click the type icon on the date column header and select 'Date' to ensure Power BI recognizes it for time intelligence functions.
3	**Create a Date Table**	In Power BI Desktop, go to Modeling > New Table and use CALENDAR () or CALENDARAUTO () DAX to generate a proper date dimension for time intelligence.
4	**Standardize region and category labels**	Use Transform > Replace Values or Add Column > Conditional Column to normalize inconsistent text values in the Region and Product Category fields.
5	**Encode customer fields**	Map occupation and gender fields to clean, consistent labels using Conditional Columns in Power Query (e.g., 'M' → 'Male', 'F' → 'Female', null → 'NA').

6.4 Verify Data Types

Purpose: Confirm every column is assigned the correct data type to ensure DAX measures calculate accurately and visuals render correctly.

Step	Action	Details
1	**Review column type icons**	In the Power Query Editor, every column header displays a type icon: ABC (text), 123 (whole number), 1.2 (decimal), or calendar (date). Review all columns.
2	**Fix text-stored numbers**	If revenue or quantity columns show ABC icons, click the icon and select 'Decimal Number' or 'Whole Number' to convert them properly.
3	**Validate date columns**	Date columns must show the calendar icon. If not, change the type to 'Date'. Verify by confirming YEAR(), MONTH(), and QUARTER() functions work in DAX.
4	**Check for error rows**	Enable Column Quality (View tab) and inspect any columns showing Error %. Click 'Errors' to filter and review problematic rows, then decide to remove or correct them.
5	**Run Column Profile audit**	Enable 'Column Profile' under the View tab and set it to profile the entire dataset. Review Min, Max, Distinct Count, and Value Distribution for all key columns before closing.

7. Dashboard Walkthrough

The Sales Performance Dashboard is structured across seven report pages, accessible via the left-hand navigation panel in Power BI. A Region Filter slicer (Australia, Canada, Central, France, Germany, Northeast, and more) and a Date Range selector (Jan 2015 – Mar 2017) are available at the top of each page for cross-page filtering.

All monetary values are reported in USD. Data source: Sales Database. Last refreshed: May 21, 2024 at 10:30 AM.

7.1 Overview

The Overview page provides the executive summary of Well-Up's commercial performance. It surfaces the four headline KPIs and six key charts covering revenue, profit, regional distribution, customer segments, product categories, and time trends.

KPI Summary Cards

$24.91M	$10.46M	84K	1,828
Total Revenue	Total Profit	Total Qty Sold	Total Returns

Additional context: Profit Margin is 42.0% | Overall Revenue Growth from Jan 2015 to Mar 2017: 209.4%

Charts on This Page

Chart	Visual Type	Key Finding
Revenue by Region	Horizontal Bar Chart	Australia leads with $6.91M. Southwest ($4.32M) and Northwest ($2.81M) follow. France is lowest at $1.58M.
Occupation by Customer Patronage	Horizontal Bar Chart	Professionals generate the most revenue at $8.50M. Manual workers contribute the least at $2.50M.
Revenue by Product Categories	Column Chart	Category 1 dominates at $24.27M. Categories 2–4 are significantly lower ($0.12M–$0.32M).
Revenue by Customer Gender	Donut Chart	Male customers lead at 50.23% ($12.52M). Female customers contribute 49.14% ($12.24M). NA is 0.63%.
Revenue by Year, Quarter & Month	Line/Area Chart	Revenue grew from $0.53M in Jan 2015 to $1.83M in Mar 2017, representing 209.4% overall growth.
Key Insights Panel	Static Summary	Highlights: 209.4% revenue growth, 42.0% profit margin, and Australia as top region by revenue.

7.2 Sales Analysis

The Sales Analysis page provides a deeper breakdown of sales performance by region, time period, and transaction volume, enabling the sales team to identify peak periods and regional trends.

Visual	Type	Purpose
Revenue trend over time	Line Chart	Shows month-by-month revenue movement to identify seasonal peaks and troughs.
Sales by region comparison	Bar/Matrix	Compares revenue contribution across all regions side by side.
Transaction volume analysis	Column Chart	Tracks the number of transactions per period to correlate with revenue changes.
Regional revenue breakdown	Table / Matrix	Detailed tabular view of sales figures filterable by region and date range.

7.3 Product Analysis

The Product Analysis page focuses on category-level performance, helping procurement and marketing teams understand which products drive the most revenue and profitability.

Visual	Type	Purpose
Revenue by Product Category	Column Chart	Ranks all product categories by total revenue generated across the date range.
Profit by Product Category	Bar Chart	Shows profit contribution per category to identify high-margin vs. low-margin lines.
Return Rate by Product	KPI / Bar	Highlights categories with elevated return rates that may signal quality or expectation issues.
Category Revenue Trend	Line Chart	Tracks category-level revenue over time to spot growth or decline patterns.

7.4 Customer Insights

The Customer Insights page segments revenue and purchasing behavior by customer occupation and gender, providing the marketing and CRM teams with data to support targeting strategies.

Visual	Type	Key Finding

Revenue by Occupation	Bar Chart	Professionals ($8.50M) and Skilled Manual ($5.40M) are the top two customer groups by revenue.
Revenue by Gender	Donut Chart	Near-equal split: Male 50.23%, Female 49.14%, NA 0.63% — indicating a balanced customer base.
Customer patronage trend	Line Chart	Shows how each occupation group's purchasing has trended over the Jan 2015–Mar 2017 period.
Occupation vs. Profit Margin	Matrix / Table	Cross-references occupation segment against profit margin to find highest-value customer groups.

7.5 Time Intelligence

The Time Intelligence page leverages DAX time functions to deliver period comparisons, cumulative totals, and growth rate calculations — critical for executive trend reporting.

Measure / Visual	DAX Function Used	Purpose
Year-over-Year Revenue Growth	SAMEPERIODLASTYEAR()	Compares current period revenue against the same period in the prior year.
Month-to-Date Revenue	DATESMTD()	Shows cumulative revenue from the start of the current month to the selected date.
Quarter-to-Date Revenue	DATESQTD()	Shows cumulative revenue for the current quarter up to the selected date.
Year-to-Date Revenue	DATESYTD()	Tracks cumulative annual revenue from Jan 1 to the current date in the filter context.
Revenue Growth % (Overall)	Custom DAX Measure	Overall growth from Jan 2015 to Mar 2017 calculated at 209.4% — shown on Overview page.

7.6 Geography

The Geography page uses map-based visuals to present revenue distribution spatially, making it easy for regional managers to identify high-performing and underperforming territories.

Visual	Type	Key Finding
Revenue by Region (Map)	Filled Map / Bubble Map	Australia ($6.91M) and Southwest ($4.32M) visually dominate the revenue map.
Region Filter interaction	Slicer + Map	Clicking any region button dynamically updates all visuals to show region-specific data.
Top Region callout	KPI Card	Australia is highlighted as the top region by revenue in the Key Insights panel on the Overview page.
Region comparison table	Matrix Table	Presents all regions with their revenue, profit, qty sold, and return metrics side-by-side.

7.7 DAX Metrics

The DAX Metrics page surfaces all calculated measures used across the dashboard, providing full analytical transparency. It serves as a reference for data analysts and report consumers who want to understand the logic behind each KPI.

Measure	DAX Formula (Summary)	Output
Total Revenue	SUM(Sales[Revenue])	$24.91M
Total Profit	SUM(Sales[Profit])	$10.46M
Profit Margin %	DIVIDE([Total Profit], [Total Revenue])	42.0%
Total Quantity Sold	SUM(Sales[Quantity])	84K units
Total Returns	SUM(Sales[Returns])	1,828 items
Revenue Growth %	([Revenue CY] - [Revenue PY]) / [Revenue PY]	209.4%

8. Key Insights & Recommendations

Based on the dashboard analysis covering January 2015 to March 2017, the following insights and recommended actions are identified:

#	Insight	Recommended Action

1	Revenue grew 209.4% from Jan 2015 to Mar 2017.	Investigate the primary drivers of this growth (region, product, or customer) and replicate the strategy going forward.
2	Australia is the top revenue region at $6.91M — over 1.5× the next region (Southwest at $4.32M).	Deepen investment in Australia through expanded product offerings and targeted promotions. Analyze what makes it outperform.
3	Profit Margin stands at 42.0%, indicating strong profitability.	Benchmark this margin against industry standards and protect it by monitoring cost creep in lower-margin regions.
4	Professionals generate the highest revenue at $8.50M — 3.4× more than Manual workers ($2.50M).	Prioritize professional-segment marketing campaigns and tailor product bundles to this high-value customer group.
5	Male and Female customers are nearly equal in revenue contribution (50.23% vs. 49.14%).	Maintain gender-balanced marketing. Use this parity to test gender-specific campaigns without risking revenue concentration.
6	Product Category 1 dominates at $24.27M — far ahead of Categories 2–4 (under $0.33M combined).	Investigate whether Categories 2–4 need re-positioning, pricing review, or marketing investment to grow their contribution.
7	1,828 total returns recorded — a metric requiring careful monitoring.	Analyze returns by product category and region to identify root causes. A high return rate in specific SKUs may indicate quality issues.
8	France ($1.58M) and Germany ($1.73M) are the two lowest-revenue regions.	Conduct market research in France and Germany to understand adoption barriers. Consider localized pricing or distribution strategies.

End of Documentation — Well-Up | Sales Performance Dashboard Project